Standard Operating Procedure (SOP)

Preparation of Giemsa Stock Solution

1. Purpose

To prepare and use Giemsa stain for microscopic staining of blood films and parasites (e.g., Plasmodium).

2. Principle

Giemsa is a Romanowsky-type stain containing Azure, Eosin, and Methylene blue. It differentially stains blood cells and parasites, aiding morphological identification.

A. Preparation of Giemsa Stock Solution (3.8%)

Reagents:
• Giemsa powder – 3.8 g
• Glycerol (analytical grade) – 250 mL
• Methanol (absolute) – 250 mL

Procedure:
1. Weigh 3.8 g of Giemsa powder.
2. Add it to 250 mL of glycerol in a glass-stoppered bottle.
3. Warm gently in a water bath at 55–60 °C for 1 hour, stirring intermittently until the dye dissolves completely.
4. Add 250 mL of absolute methanol, mix thoroughly.
5. Label as 'Giemsa Stock Solution (3.8%)', including date and preparer's initials.
6. Store in a dark bottle at room temperature.
7. Allow to mature for at least 1 week before use.

B. Preparation of Working Giemsa Solution

For thick and thin blood films, dilute 1 part of stock solution with 9 parts of buffered water (pH 7.2). That is, 10% v/v solution — e.g., 10 mL stock + 90 mL buffer.

Buffer Preparation (pH 7.2)
Mix 6.5 mL of KH■PO■ (0.067 M) with 43.5 mL of Na■HPO■ (0.067 M), and make up to 100 mL with distilled water.

C. Quality Control

• Use known positive and negative control slides.
• Store stain in tightly capped bottles away from sunlight.
• Replace stock if precipitates or color change occurs.